

CS4234: Optimization Algorithms
Semester 1, AY 2026–27
School of Computing, NUS

Assignment 1

Deadline: 18 September (Friday), 23:59

Instructions.

- This assignment contains three questions and carries a total of 30 marks.
 - Submit a single PDF file through **Canvas** → **Assignments** → **Assignment 1** before the deadline.
 - Give complete, mathematically precise arguments. Clearly state the algorithm, its running time, and the approximation guarantee. A correct answer without adequate justification may receive only partial credit.
 - Unless a question states otherwise, all input numbers are nonnegative and have polynomial-size encodings. For a randomized algorithm, it is sufficient to prove the requested guarantee in expectation.
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Question 1: A Collision-Free Sensor Upload [10 marks]

A network of remote environmental sensors shares a collection of wireless gateways. During a short emergency-upload window, the system must choose a subset of the sensors to transmit. Every gateway j can hear exactly $k \geq 2$ distinct sensors; let B_j denote the set of such sensors for gateway j . (Note the sets B_j may overlap, i.e., for any two gateways $j \neq j'$, $B_j \cap B_{j'}$ need not be empty.) Gateway j successfully decodes a packet if *exactly one* sensor in B_j transmits. If none transmits, the gateway receives no packet; if two or more transmit, their packets collide. Successfully serving gateway j has importance $p_j \geq 0$.

The objective is to choose a set of transmitting sensors that maximizes the total importance of the gateways that successfully decode a packet. Let OPT denote the maximum possible importance.

Design a randomized polynomial-time algorithm whose expected value is at least OPT/e . Your solution should determine and justify the transmission probability used by the algorithm, prove the approximation guarantee, and state the running time.

Question 2: Planning a Biodiversity Survey [10 marks]

A conservation agency is choosing field-survey routes. Let U be the set of species that might be observed, and let $a_u \geq 0$ be the conservation importance of species $u \in U$. Suppose there are m routes. Route i costs c_i , observes a subset $S_i \subseteq U$, and may overlap with other routes. The agency has budget B , and every route satisfies $0 < c_i \leq B$. A species contributes its importance at most once, regardless of how many chosen routes observe it.

For a set A of routes, define its *value* as follows:

$$F(A) = \sum_{u \in \bigcup_{i \in A} S_i} a_u.$$

The goal is to maximize $F(A)$ subject to $\sum_{i \in A} c_i \leq B$.

Design a deterministic polynomial-time algorithm that returns a feasible subset of routes with value at least $\frac{1}{2} \left(1 - \frac{1}{e}\right) \text{OPT}$, where OPT is the value of an optimal budget-feasible subset. Prove the approximation guarantee and state the running time of your algorithm.

Question 3: Emergency Communication after a Storm [10 marks]

Following a coastal storm, an emergency team may activate temporary communication relays at candidate sites. Activating site i costs $c_i \geq 0$. A community j can be reached only from sites in a specified set $N(j)$. If no activated relay reaches community j , the team incurs a penalty $p_j \geq 0$ for arranging a slower alternative. Each community can be reached from at most f candidate sites, where $f \geq 1$ is given; that is, $|N(j)| \leq f$ for every community j .

The goal is to minimize

$$\sum_{i \text{ activated}} c_i + \sum_{j \text{ not reached by an activated site}} p_j.$$

- (a) Formulate this problem as an integer linear program and write its natural LP relaxation. [3 marks]
- (b) Using an optimal solution to the LP relaxation, give a deterministic polynomial-time rounding algorithm. [3 marks]
- (c) Prove that your algorithm is an $(f + 1)$ -approximation algorithm. [4 marks]